

NCP-AIN AI Fabric Pro Mega Lab I Print A3 Landscape

Underlay Port / IP Mapping

LEAF1 E1 10.0.0.0/31 <-> SPINE1 E1 10.0.0.1/31
LEAF1 E2 10.0.0.12/31 <-> SPINE2 E1 10.0.0.13/31
LEAF2 E1 10.0.0.2/31 <-> SPINE1 E2 10.0.0.3/31
LEAF2 E2 10.0.0.14/31 <-> SPINE2 E2 10.0.0.15/31
LEAF3 E1 10.0.0.4/31 <-> SPINE1 E3 10.0.0.5/31
LEAF3 E2 10.0.0.16/31 <-> SPINE2 E3 10.0.0.17/31
LEAF4 E1 10.0.0.6/31 <-> SPINE1 E4 10.0.0.7/31
LEAF4 E2 10.0.0.18/31 <-> SPINE2 E4 10.0.0.19/31
BL1 E1 10.0.0.8/31 <-> SPINE1 E5 10.0.0.9/31
BL1 E2 10.0.0.20/31 <-> SPINE2 E5 10.0.0.21/31
BL2 E1 10.0.0.10/31 <-> SPINE1 E6 10.0.0.11/31
BL2 E2 10.0.0.22/31 <-> SPINE2 E6 10.0.0.23/31
RR1 E1 10.0.100.1/31 <-> SPINE1 E10 10.0.100.0/31
RR1 E2 10.0.100.3/31 <-> SPINE2 E10 10.0.100.2/31
RR2 E1 10.0.100.5/31 <-> SPINE1 E11 10.0.100.4/31
RR2 E2 10.0.100.7/31 <-> SPINE2 E11 10.0.100.6/31

Fabric Roles / Loopbacks

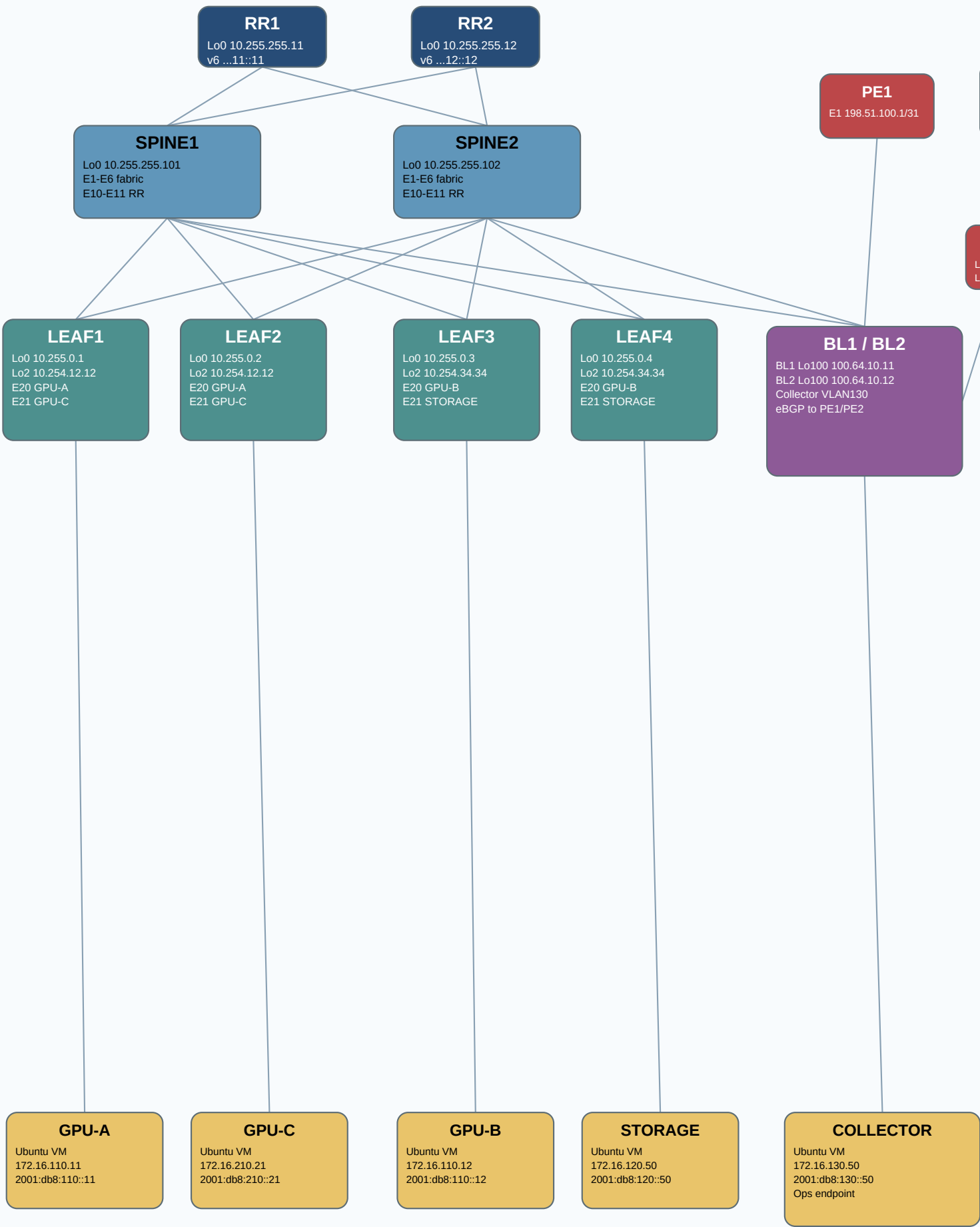
RR1 Lo0 10.255.255.11 | RR2 Lo0 10.255.255.12
SPINE1 Lo0 10.255.255.101 | SPINE2 Lo0 10.255.255.102
LEAF1 Lo2 10.254.12.12 | LEAF2 Lo2 10.254.12.12
LEAF3 Lo2 10.254.34.34 | LEAF4 Lo2 10.254.34.34
BL1 Lo1 10.254.0.11 | BL2 Lo1 10.254.0.12
BL1/BL2 border VTEP Lo2 10.254.78.78
RR1/RR2 are EVPN route reflectors
Spines carry underlay transit only

Host / Border Ports

GPU-A eth0->LEAF1 E20 | eth1->LEAF2 E20 | Po10
GPU-C eth0->LEAF1 E21 | eth1->LEAF2 E21 | Po11
GPU-B eth0->LEAF3 E20 | eth1->LEAF4 E20 | Po20
STORAGE eth0->LEAF3 E21 | eth1->LEAF4 E21 | Po21
COLLECTOR eth0->BL1 E20 | eth1->BL2 E20 | Po30
BL1 E30 198.51.100.0/31 <-> PE1 E1 198.51.100.1/31
BL2 E30 198.51.100.2/31 <-> PE2 E1 198.51.100.3/31

Physical Topology

Exact port/IP assignments are in the adjacent tables



Dual-Stack VLAN / VNI / GW

VLAN110 | VNI10110 | VRF AI_FABRIC
IPv4 172.16.110.0/24 | GW 172.16.110.1
IPv6 2001:db8:110::/64 | GW 2001:db8:110::1
VLAN120 | VNI10120 | VRF AI_FABRIC
IPv4 172.16.120.0/24 | GW 172.16.120.1
IPv6 2001:db8:120::/64 | GW 2001:db8:120::1
VLAN130 | VNI10130 | VRF AI_SERVICES
IPv4 172.16.130.0/24 | GW 172.16.130.1
IPv6 2001:db8:130::/64 | GW 2001:db8:130::1
VLAN210 | VNI10210 | VRF TENANT_B
IPv4 172.16.210.0/24 | GW 172.16.210.1
IPv6 2001:db8:210::/64 | GW 2001:db8:210::1

Endpoint Addressing

GPU-A 172.16.110.11 | 2001:db8:110::11
GPU-B 172.16.110.12 | 2001:db8:110::12
GPU-C 172.16.210.21 | 2001:db8:210::21
STORAGE 172.16.120.50 | 2001:db8:120::50
COLLECTOR 172.16.130.50 | 2001:db8:130::50
REMOTE 203.0.113.10/32 | 2001:db8:203:113::10

Notes

GPU nodes are Ubuntu VMs.
Use tables as source of truth for exact mapping.
Workbook covers config, validation, and faults.
IB behavior is conceptual in EVE-NG, not ASIC-accurate.
RoCE/EVPN/VXLAN/routing workflows are practice-focused.